

GRAPH-BASED DETECTION

Network Flow Telemetry
(IoT and Enterprise Captures)
Raw flows → edge feature vectors

Communication Graph

Nodes = IPv4 hosts | Edges = directed network flows
Inductive split: message-passing edges = training only

Detector

Edge-centric inductive GNN (E-GraphSAGE family)
Jumping-knowledge fusion + target-edge scoring

Detector Outputs

Edge Classification
Class logits
Classification confidence

Edge Reconstruction
Benign reconstruction error
Normalized anomaly score

Hybrid Confidence: classification confidence + normalized anomaly score

Prediction | Subgraph Signature | Confidence

Evidence Store
Built Only from Historical Traffic

RETRIEVAL-AUGMENTED EXPLANATION (RAG)

Confidence Check
Classifier confidence $< \tau$
AND hybrid score $< \tau_h$?

Confident (NO)
Continue to RAG Stage

Uncertain (YES)
Escalate: No LLM Explanation
→ Analyst Review

Structured Evidence Set (Attribution & Context)

Flow Details

Influential Features
Gradient × Input attribution

Local Subgraph Context
Local Graph Stats

Retrieved Similar Subgraphs
Retrieved Precedents

Retrieval & Ranking

Normalized edge embedding

Dual index (benign / attack)

MMR diversity re-rank

GROUNDING VERIFICATION

Explanation Generator

Prompt Construction

Triple Serialization

Structured JSON Prompt

Local language model

Structured Output (JSON)

Predicted Class | Rationale
Confidence Level | Counterfactual
Evidence List

Grounding Audit (2 gates + 3 checks)

S1 Label Consistency | H1 Feature Grounding | H3 Precedent-Label Check
S2 Confidence Range | H2 Numeric Faithfulness

Grounded

Explanation Withheld

Verified Explanation

Security Analyst

Analyst Review

Text-Quality Metrics (reported, not used for gating)

ROUGE-L | Self-BLEU | BERTScore | Length

Output